# Ali Abbasi

Fullstack AI Engineer

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## Summary

**Fullstack AI Engineer** and hands-on technical lead with 5+ years of software development experience, originally as a fullstack developer and now focused on shipping production GenAI systems. I own LLM-based products end-to-end: from solution design and agentic RAG architecture, to fine-tuning, evaluation, and reliable deployment on AWS with containerized services, CI/CD, and production monitoring.

I combine applied research experience with delivery ownership, leading small teams and translating ambiguous stakeholder needs into production AI systems. My background also covers Reinforcement Learning, Computer Vision, and published applied research.

## Experience

### AI Engineer (Team Lead, Part-time, Remote)

*London, UK*

Primentoring.AI

*Nov 2025 - Present*

- Lead a 2-engineer team end-to-end, owning solution architecture, feasibility research, project health reporting, and stakeholder alignment across product, QA, and software teams.
- Architected Dana AI, a real-time multimodal agent (3D talking head, voice, text) over WebSocket: OmniTalker for lip-sync (LoRA fine-tuned per mentor on AWS SageMaker), GPT-4o Transcribe ASR with vocabulary hints, and GPT-4.1 fine-tuned on the OpenAI platform for mentor tone and persona, at ~2.5s time-to-first video response frame.
- Designed a production agentic RAG stack on self-managed Weaviate (AWS EC2) with ~900 ms retrieval latency, hybrid search, query expansion, reranking, function-calling agents, task decomposition, and conversational + long-term memory via LangGraph (~87% top-5 retrieval precision); reused as the backbone of an LLM Customer Support Chatbot over company documents.
- Built a Public Data Aggregation feature that extracts mentor signals from 8 platforms in parallel, completing each collection run in ~3 minutes while respecting each service's ToS limits and GDPR.
- Built a RAGAS-based evaluation suite with 100 curated mentor-domain queries, plus guardrails, PII filtering, confidence scoring, human-in-the-loop handoff, prompt/version tracking, and a closed-loop Unanswered Pool that routes low-confidence queries to mentors and ingests responses back into the knowledge base.
- Built a recommender system that extracts behavioral signals (skills, engagement, talk-time, emotion) from mentoring sessions to match mentors to mentees.
- Operationalized all AI services as Django/Celery microservices on EC2 with Redis, PostgreSQL, ETL pipelines, GitHub Actions CI/CD reducing deployment time to ~1.5 minutes, and Prometheus + Grafana monitoring.

### Machine Learning Engineer | Fullstack Developer

*Turin, Italy*

Nimbus SRL

*Feb 2022 - Present*

- Built and shipped an internal LLM-powered support assistant (LangChain RAG over a LoRA fine-tuned Llama-3-8B on AWS SageMaker, grounded in proprietary docs), now serving 1,200+ weekly queries.
- Cut new-engineer onboarding from weeks to 2-3 days by exposing the fine-tuned model through an internal chat interface backed by the company knowledge base.
- Designed a PPO-based Reinforcement Learning controller to stabilize VTOL autopilot under degraded weather, improving recovery success by 5% in adverse-weather flight test simulations.
- Led a 4-engineer team to deliver a cross-platform fleet monitoring system on AWS, providing 2-second-interval telemetry for 150+ UAVs across Europe.
- Modernized the software stack (React Native, Django REST Framework, CI/CD, containerized deployments), saving ~€50,000/year by reducing the engineering capacity required to maintain and evolve the platform.
- Contributed to the Charger Box project, automating drone repositioning for autonomous charging and achieving 98% successful docking.

### Machine Learning Engineer | Fullstack Developer

*Turin, Italy*

Reefilla SRL

*Sep 2021 - Feb 2022*

- Designed and deployed a recommender model that predicts the optimal timing for electric power-bank requests, improving delivery responsiveness by ~18%.
- Co-developed a cross-platform application (React Native + Django) for EV charging management, growing the user base to 800+ active users within 2021.

## Earlier Experience

**Backend Developer** | Bardal Startup | *Tehran, Iran* | *Jul 2019 - Jan 2020* | ***Django, DRF,REST API, Docker, Kubernetes, PostgreSQL***

**Frontend Developer** | Spinzer Startup | *Tehran, Iran* | *Nov 2018 - Jun 2019* | ***ReactJS, TypeScript, Redux, HTML/CSS***

**Backend Developer (Internship)** | Saaed Co. | *Arak, Iran* | *Jun 2018 - Oct 2018* | ***.Net, REST API, HTML/CSS, JS***

# Projects

### Local Image Studio: Private AI Image Generation and Editing

Personal Project *Oct 2026 - Nov 2026*

- Self-hosted, open-source studio for creating and editing images on your own GPU with FLUX.2 Klein 9B, SAM 3 and ComfyUI: MLX on Apple Silicon, GGUF with CUDA on NVIDIA GPUs from 6 GB of VRAM. One-command installer and Docker Compose deployment.
- Text-to-image, instruction-based and masked editing with SAM 3 automatic masks and face protection, close-up refinement, head and person swaps, and image-to-video with LTX-2.3. Jobs survive restarts, and every image keeps a reproducible recipe.
- Patched the MLX pipeline to free the text encoder and transformer between stages, cutting an 8-bit 1024×1024 image from 184 s to 76 s with identical pixels; the engine starts on demand and unloads after 10 idle minutes.
- Installable phone PWA over HTTPS with a name-constrained home CA, PIN sign-in, photo sharing from the gallery and an offline library.

### Vector DB Manager for Weaviate and Qdrant

Personal Project *Mar 2025*

- Next.js/TypeScript dashboard for managing Weaviate and Qdrant vector databases: CRUD over collections and objects, semantic and hybrid search, live metrics, and one-command Docker deployment.

### Reconstructing Human Gaze Behavior in Stroop Test Using IRL and EEG Data

Alabama University & Mississippi State University & Polytechnic University of Turin *Jul 2024 - Dec 2024*

- Developed an Inverse Reinforcement Learning (IRL) model to reconstruct human gaze behavior in a Stroop test.
- Fine-tuned Detectron2 FPN model for feature extraction of low res and high res Stroop tasks.
- Used PPO and GAE inside GAIL algorithm.
- Analyzed the impact of cognitive load on gaze patterns and decision-making processes.

### COVID-19 Pandemic Network Modelling Using GSM Mobility Data and SIR Model

Polytechnic University of Turin *Jan 2024 - Mar 2024*

- Developed a network model to simulate the spread of COVID-19 using GSM mobility data and the SIR model on weekly basis.
- Used library such as NetworkX, Pandas, and NumPy for data manipulation and analysis.
- Used CUDA for efficient HPC of the SIR model on large-scale networks.
- Analyzed the impact of mobility patterns on the transmission dynamics of the virus.
- Implemented various scenarios to assess the effectiveness of different intervention strategies.

### Distributed Learning on CIFAR-100 dataset: Data-Parallelism Study

Polytechnic University of Turin *Nov 2023 - Dec 2023*

- Implemented distributed machine learning algorithms, LocalSGD, Distributed SGD, SlowMo, and LocalAdaScale over multiple GPU nodes.
- Compared the performance of algorithms on CIFAR-100 dataset in terms of convergence speed, accuracy, and communication overhead.

### TFLite Triggerer Words Recognition Model

Polytechnic University of Turin *Oct 2023*

- Developed a model using TFLite and LSTM architecture on Arduino to detect the triggerer word under low voltage and low memory constraints.

# Skills

**Programming Languages:**
Python, C++, C#, JavaScript, TypeScript, SQL

**GenAI & LLM Engineering:**
LangGraph, LangChain, Agentic RAG, Hybrid Search, Reranking, Function Calling, Task Decomposition, Conversational & Long-term Memory, MCP, LoRA Fine-tuning, OpenAI Fine-tuning Platform, RAGAS, Guardrails, PII Filtering, Prompt Versioning

**Machine Learning:**
PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, Reinforcement Learning, Computer Vision, NLP, WandB, TensorBoard

**Data & Storage:**
PostgreSQL, Redis, Weaviate, MySQL, SQLite, ETL Pipelines

**Backend & APIs:**
Django, Django REST Framework, Celery, REST, WebSocket, Microservices, Pytest

**DevOps & Cloud:**
AWS (EC2, SageMaker, S3), Azure, Docker, Kubernetes, GitHub Actions, GitLab CI, Nginx, Prometheus, Grafana, Linux

**Frontend & Mobile:**
React.js, React Native, Next.js, Redux, Tailwind CSS

**Leadership & Collaboration:**
Technical Leadership, Solution Architecture, Stakeholder Management, Feasibility Analysis, Mentoring, Agile/Scrum, Git, Jira, Confluence

# Education

**M.Sc. in Data Science and Engineering** *Turin, Italy*

Polytechnic University of Turin *2020 - 2024*

- **Thesis:** "Reconstructing Human Gaze Behavior in Stroop Test from EEG Data Using Inverse Reinforcement Learning"
- **Publication:** Published in Elsevier Smart Health Journal (2025) based on thesis work
- **Conference:** Also accepted at IEEE/CHASE 2025 Conference

**B.Sc. in Computer Engineering** *Arak, Iran*

Arak University *2015 - 2019*

# Publications

**Transforming Stroop task cognitive assessments with multimodal Inverse Reinforcement Learning**

Ali Abbasi, Jiaqi Gong, Soroush Korivand

Smart Health *36, p. 100567, 2025. Elsevier*

# Interests

**Reading** Absolutely hooked on business-innovation reads (e.g. Eric Ries) & self-help gems (e.g. Cal Newport).

**Football** Watch way too much Liverpool matches and play FIFA with friends.

**Learning** Always reading about new tech trends or taking online courses now focusing on AI trends.

**Night Rides** 30-40 min after-dark city spins on my bike—my go-to for a full mind & body recharge.

# Languages

**English** Professional proficiency

**Persian** Native proficiency

**References available upon request.**